

Ferran Alía Castillejos

Data science and physical engineering, UPC Barcelona

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Now In my fourth year at UPC in Barcelona, back from a summer simulating microscope optics with the Physics of Life group at TU Dresden. I want to work where physical modeling and machine learning meet.

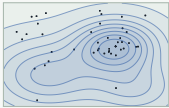
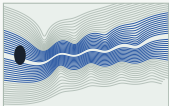
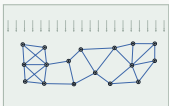
Experience

Since 2026	Contributor, Towards Data Science Long-form articles since March 2026, on simulation, high-performance computing and machine learning. Most of them come out of the projects below.
Jun – Aug 2026	Research intern, Physics of Life, TU Dresden Simulating electromagnetic propagation through specimen and objective, as virtual instrumentation for learned brightfield imaging. On the side, a GNN surrogate for multiple scattering between beads.
Jun – Aug 2025	Machine learning intern, MLCode Built an automated framework for evaluating and benchmarking retrieval-augmented generation systems.
Since 2022	Volunteer mentor Computer science and robotics workshops: Linux, 3D printing, programming.
2022 – 2023	Robotics teacher, Punt Multimèdia Taught Arduino, micro:bit, and 3D printing and modeling.

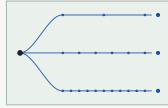
Education

2023 – 2028	BSc Data Science & Engineering, UPC In the fourth year of five. 8.75/10 average across both degrees.
2023 – 2028	BSc Physical Engineering, UPC Taken in parallel with the data science degree rather than after it, on the CFIS dual-degree program.

Selected projects

2026 	Aerodynamic optimization of a Formula 1 front wing A surrogate model searched front-wing geometries on a fixed budget of solver runs rather than calling CFD at every step. The wing came out about 15% better on lift-to-drag. Python, OpenFOAM, SLURM, MareNostrum V
2026 	A physics-informed network for vortex shedding A PINN with the Navier-Stokes residual inside its loss, trained against a high-fidelity OpenFOAM run, reproducing Kármán vortex shedding. A model fit to the data alone settles into a steady wake instead. Python, PyTorch, OpenFOAM
2026 	A graph network for multiple scattering in microscopy A GNN surrogate for the bead-bead coupling in Köhler-illuminated microscope images, fast enough to run in real time. Built at TU Dresden alongside the main internship project.

2026

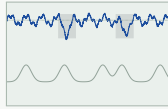


Persona adaptation in language models

A system prompt against three formats of fine-tuning data on Qwen3-4B-Instruct, all with LoRA at $r=16$, so the differences belong to the data.

Python, PyTorch, Transformers, LoRA

2025

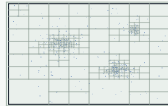


Predicting fetal distress from cardiotocography

Built with Hospital Sant Joan de Déu in a team of nine. Separate encoders over CTG traces, tabular clinical context and ultrasound, at around 0.78 AUC.

Python, PyTorch

2025

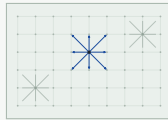


Parallel N-body simulation

Direct $O(n^2)$ summation and a Barnes-Hut $O(n \log n)$ tree code in C++ with OpenMP. At $N = 10,000$ direct summation ran about 33× faster on 112 threads than on one; the tree code never got past roughly 7×.

C++, OpenMP

2025

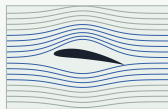


Parallel lattice Boltzmann solver

A D2Q9 solver with BGK collision and a Zou-He inlet in around 200 lines of C++: flow past a cylinder on a 400×400 lattice. Memory-bound, so how you move the data dominates the scaling.

C++, OpenMP, MareNostrum 5

2025

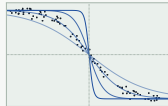


Fixed-wing bird flight simulation

A 3D Navier-Stokes solver written from scratch in NumPy, run on a bird's wing modeled in Blender and checked against OpenFOAM's icoFoam.

Python, NumPy, OpenFOAM, Blender

2024



Fermi-Dirac statistics by Monte Carlo

A Markov chain over the microstates of a fermion system, recovering the Fermi-Dirac occupancy numerically rather than analytically.

Python, NumPy

Figures are schematic illustrations of each problem, not results.

Skills

Languages

Python, C++, R, Haskell, MATLAB

Scientific

OpenFOAM, OpenMP, NumPy, PyTorch, Blender, CAD

Tools

Linux, Git

Methods

CFD, HPC & SLURM, physics-informed ML, Monte Carlo, surrogate modeling

Spoken

Catalan (native), Spanish (native), English (professional), Arabic (beginner)

Awards

Nov 2024, 2025

Participant, Datathon FME

Jun 2025

Physics in Ljubljana, summer school at the University of Ljubljana

Since 2023

Cum laude in several subjects

2023

Admitted to CFIS on a grant, to read the two degrees at once. It takes a small cohort each year by its own entrance exam in mathematics and physics

2019 - 2023

Graduated from high school with honors, and represented it at the Barcelona *Mostra de Recerca Jove*, after UPC's summer courses in algorithms and programming